Practical No. 8

Title: Exploration and Practice of SORA.

Problem Statement: Assignment and practice of SORA.

Aim: Implement and practice the use of SORA as a tool within the field of Artificial Intelligence.

Objective: Practice and implement the usage of SORA as an AI tool.

Prerequisites: Basic computer skills and internet usage. Familiarity with using web applications.

Learning Objectives:

➤ To understand the foundational concepts behind AI-driven video generation using models like SORA.
➤ To explore the architecture and workflow of SORA as a text-to-video generation model developed by OpenAI.
➤ To study the capabilities of SORA in generating realistic, coherent video content from textual prompts.
➤ To gain conceptual and observational experience through practice with SORA.

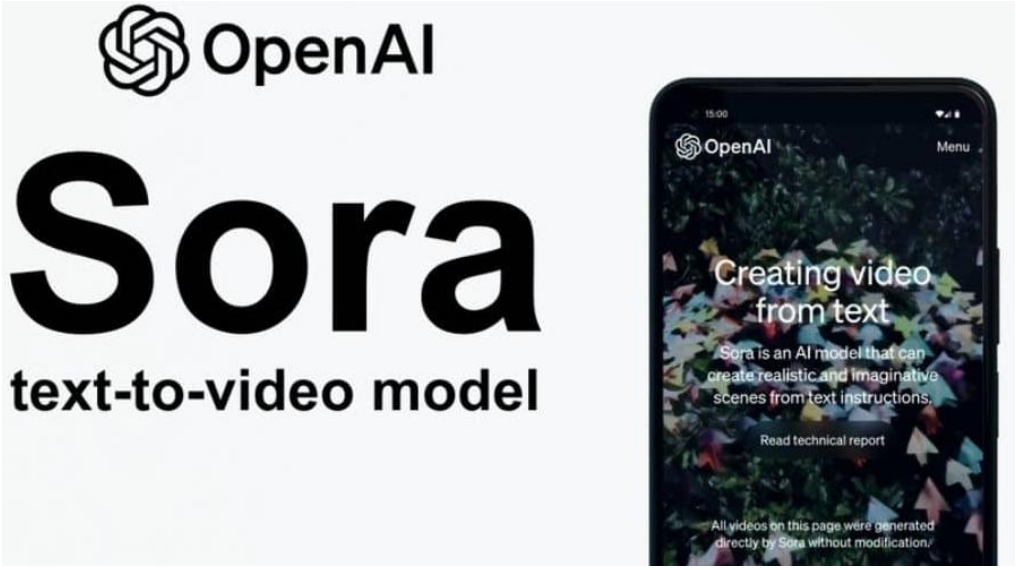
Outcomes:

✅ Understanding the basic concepts of AI-generated video and the role of models like SORA in multimedia content creation.
✅ Gaining awareness of how SORA functions as a text-to-video model, including its input-output behavior.
✅ Observing the capabilities of AI in generating realistic, coherent, and dynamic visual content from textual prompts.
✅ Developing insight into how AI-generated videos can be used in fields such as education, storytelling, marketing, and simulation.
✅ Recognizing the strengths and limitations of current video generation technology.
✅ Exploring ethical considerations around AI-generated video, including authenticity, deepfakes, and responsible use.
✅ Enhancing digital awareness and media literacy through conceptual and assignment-based engagement with SORA.
✅ Demonstrating critical thinking by evaluating AI-generated video content for coherence, quality, and relevance.

Theory:

Introduction to SORA

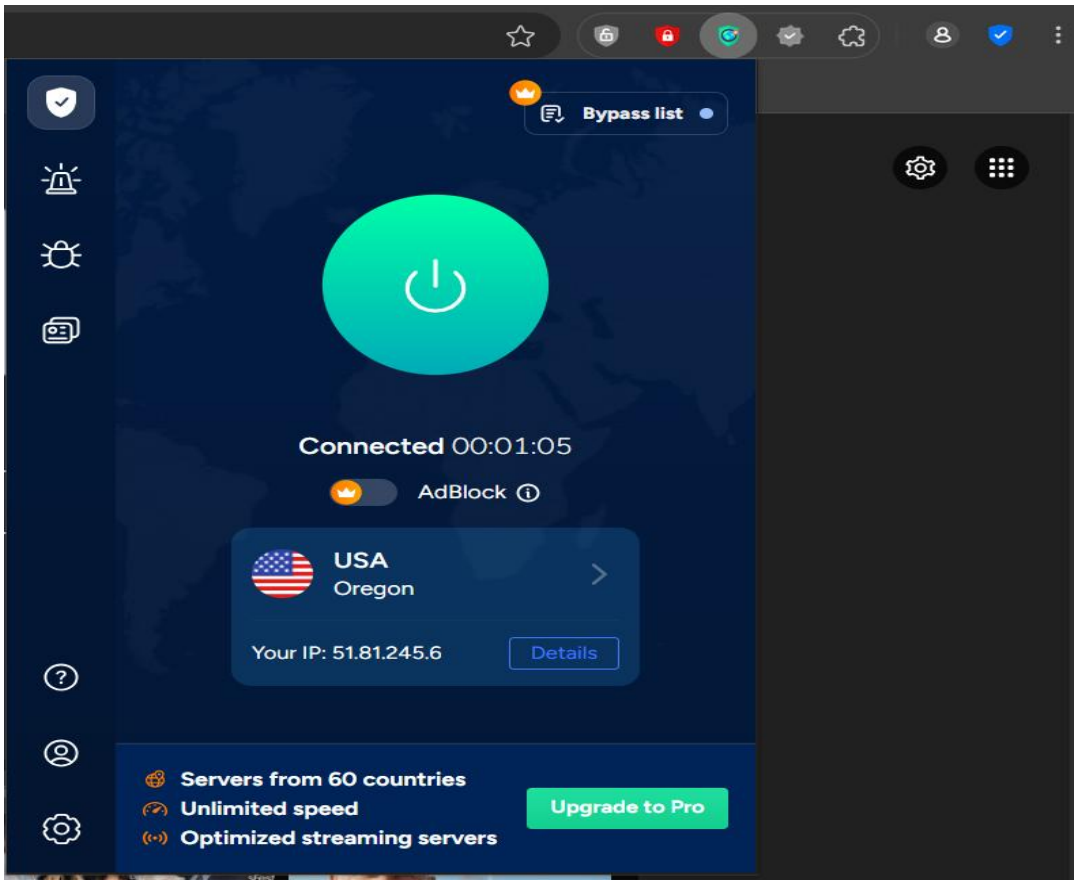
Sora is an AI-powered tool from OpenAI that turns text descriptions into short, high-quality video clips. Think of it as a virtual director that takes your words and crafts them into visuals, up to 20 seconds long, in resolutions as high as 1080p. Built on the same innovative tech as OpenAI's image generator, DALL·E, Sora understands prompts and creates realistic or imaginative scenes, complete with motion, lighting, and detail. It's like having a mini movie studio at your fingertips. Sora is a text-to-video generative AI model, released by OpenAI in February 2024. The model is trained to generate videos of realistic or imaginative scenes from text instructions and show potential in simulating the physical world.



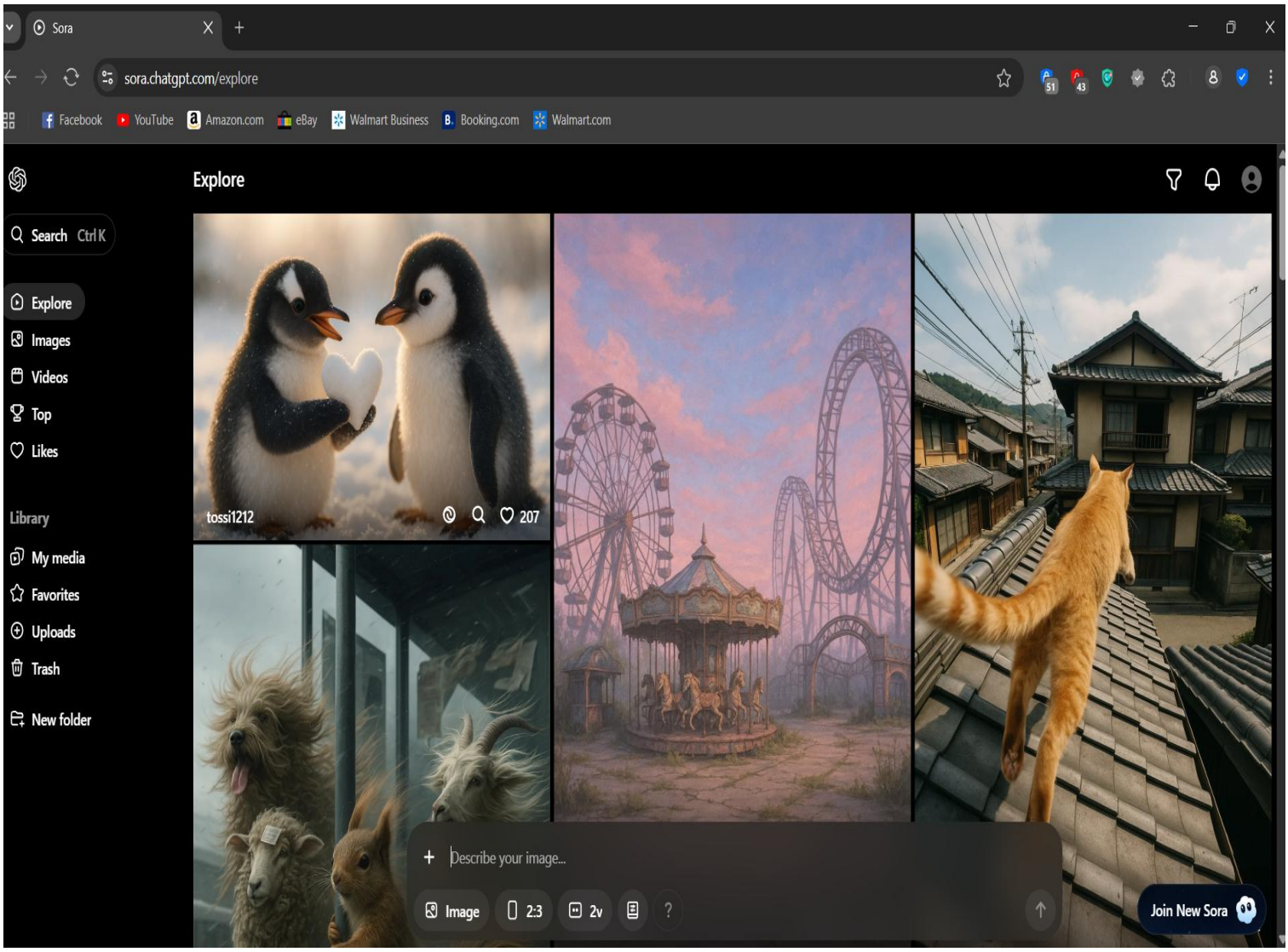
Accessing SORA

To access Sora, we need a ChatGPT account. However, as of this writing, Sora Web and Sora2 isn't available in every country yet. Only, UK and US have the access to SORA Web and Sora2. However, every country has access to SORA IOS app. SORA2 can be only accessed by invite code available to limited users via using the discord server of OpenAI. SORA2 was just recently launched, so, it is currently only available for IOS devices via App. However, in future, SORA2 will be available to all countries and will be available to all users to use without the need of invite code.

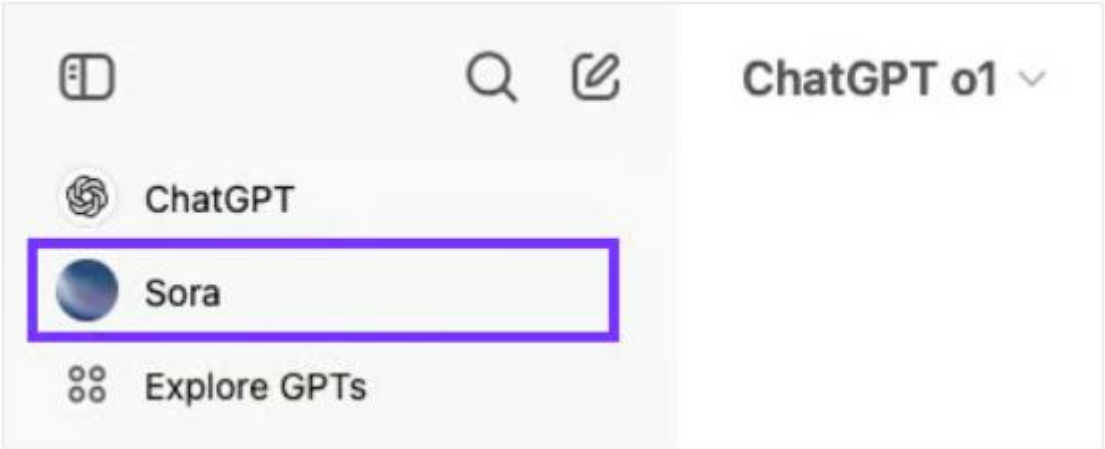
To access SORA Web from India, everyone needs to use VPN service. Browsers provide extensions and user can install VPN extension from PC. Then the user can select a manual server from a VPN either US or UK and select a city name or state name for VPN to have location in order to get access to SORA.



For other users, VPN Application can be used with location of US or UK. Other method is to use VPN based browser which has VPN location as US or UK. However, one has to use US/UK number or use Email ID for successful access to SORA. Its due to region checker of OpenAI, one cannot access by using Indian Phone Number. US/UK numbers are easily available online or via apps which support OTP and SMS receiving for Free.



To get started, we can access Sora from its dedicated website, https://sora.com/ or https://chatgpt.sora.com/ or directly from ChatGPT by clicking the Sora button on the top left:

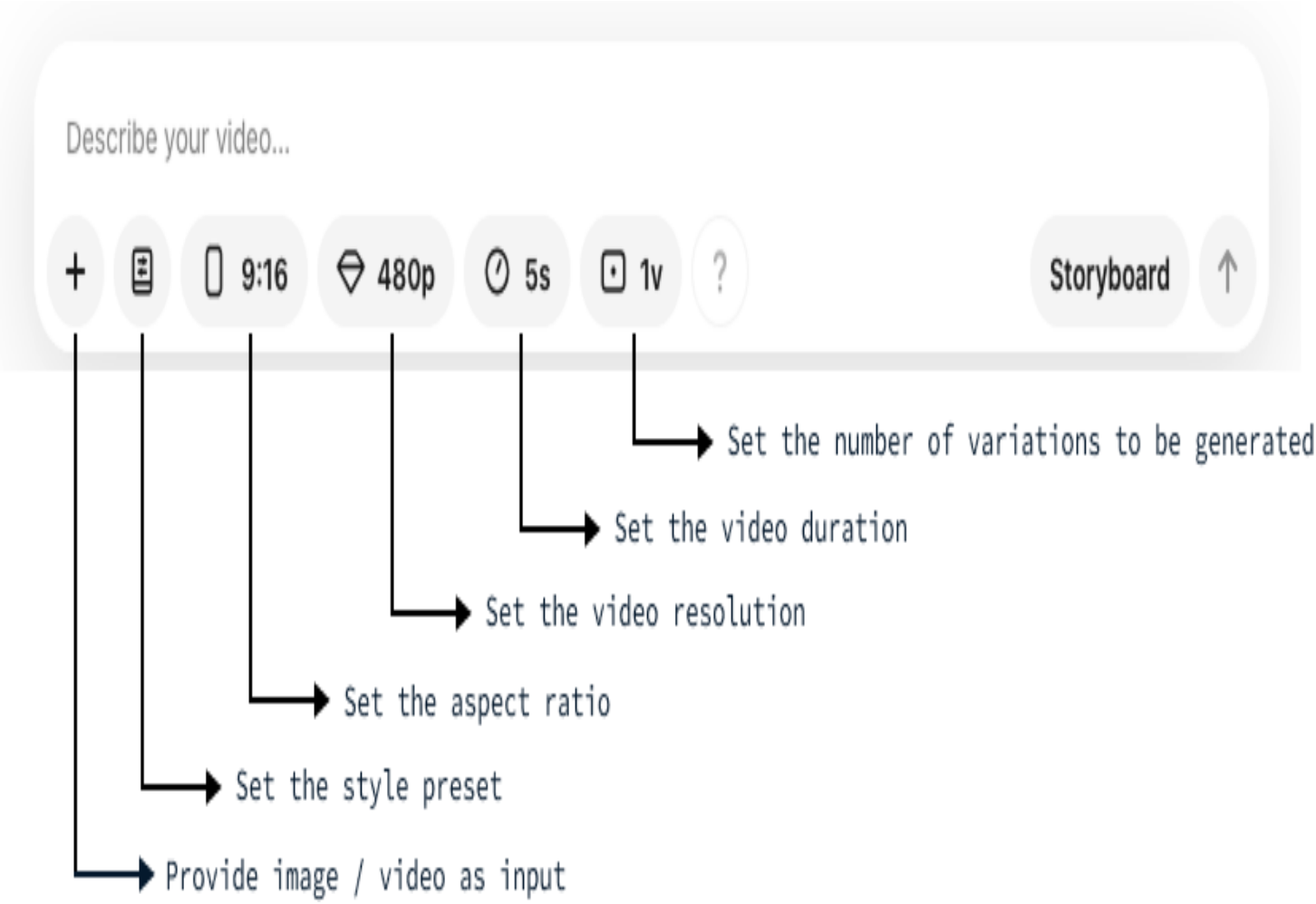


Prompt Engineering

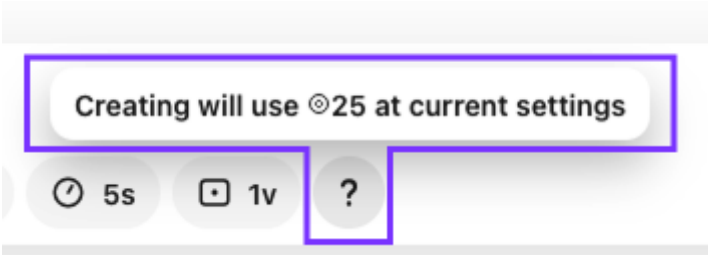
Prompt engineering refers to the process of designing and refining the input given to an AI system, particularly in the context of generative models, to achieve specific or optimized outputs. The art and science of prompt engineering involve crafting these inputs in a way that guides the model to produce the most accurate, relevant, and coherent responses possible.

Base options: Apart from the prompt, the base options available when generating a video are:

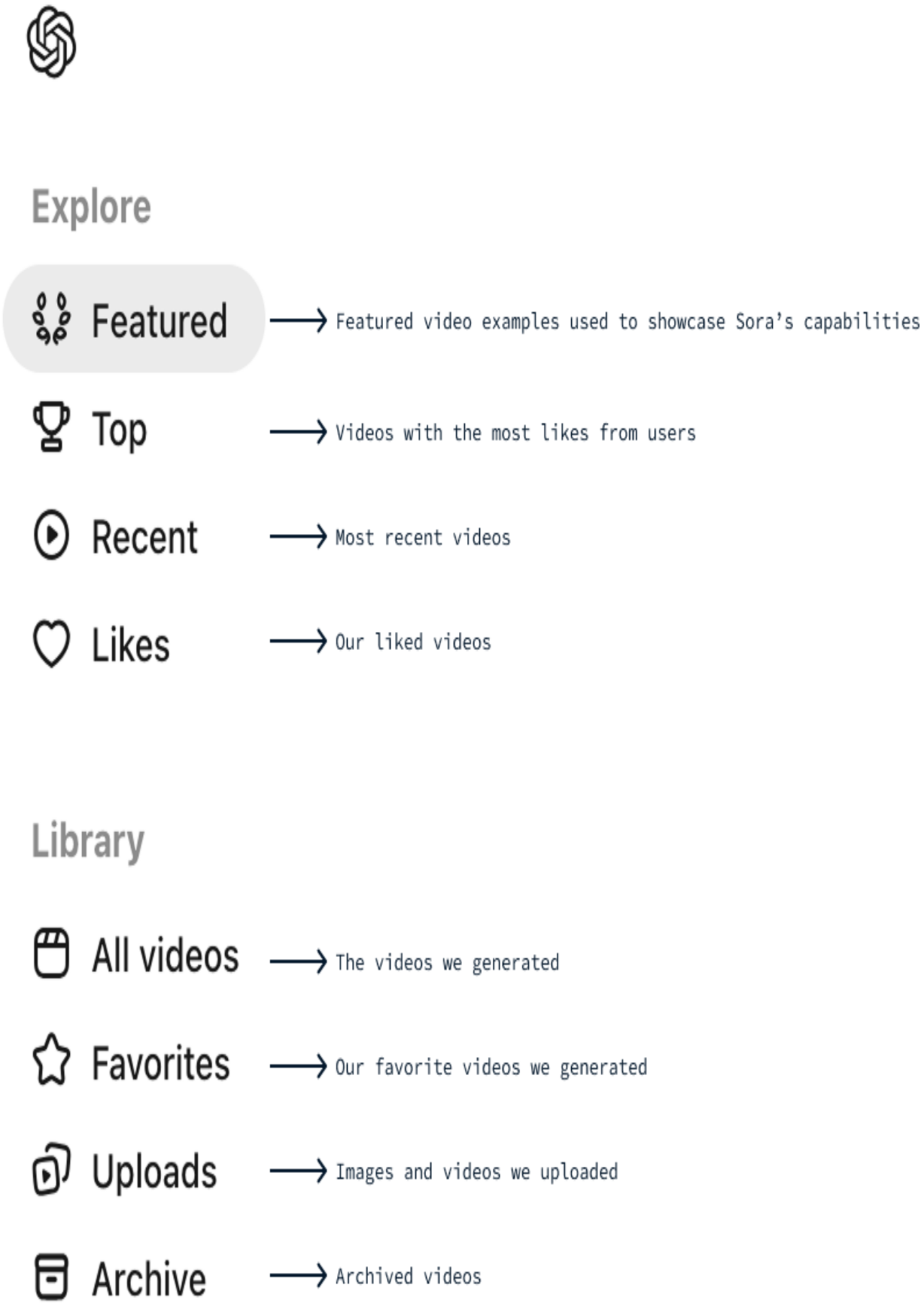
- Using an image or video to drive video generation
- Selecting a style preset
- Setting the aspect ratio: 16:9, 1:1, or 9:16
- Setting the resolution: 480p, 720p, or 1080p
- Setting the video duration: 5s, 10s, 15s, or 20s
- Select how many variations to generate: 1, 2, or 4



The last button, marked with a question mark (?), indicates how many credits will be used with the current settings when we hover it:



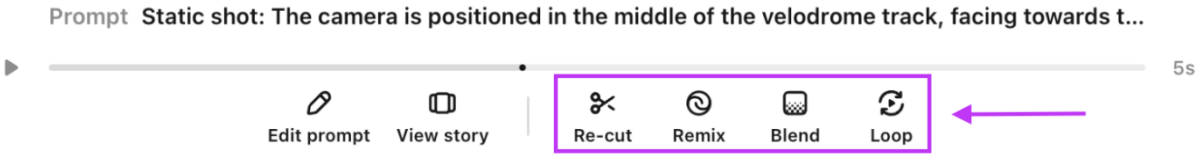
Navigating Sora's UI: Here's an overview of the Sora's menu that you can find on the left side of the home page:



If you click on the Featured section, you'll be able to see what other people created. These can be a great inspiration for your videos. To see how a video was generated, we can click on it and then hover or click on the prompt shown at the bottom of the video:



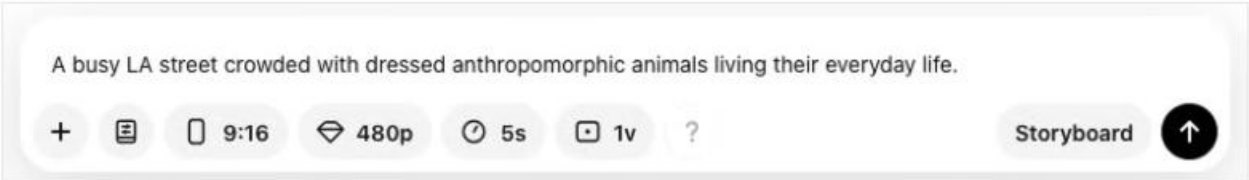
Since prompts are public, a great way to get started with Sora is by finding videos you like, checking the prompts used, and using them as a starting point for your own videos.
Once we generated a video, Sora offers us four useful tools to better control the output: re-cut, remix, blend, and loop.

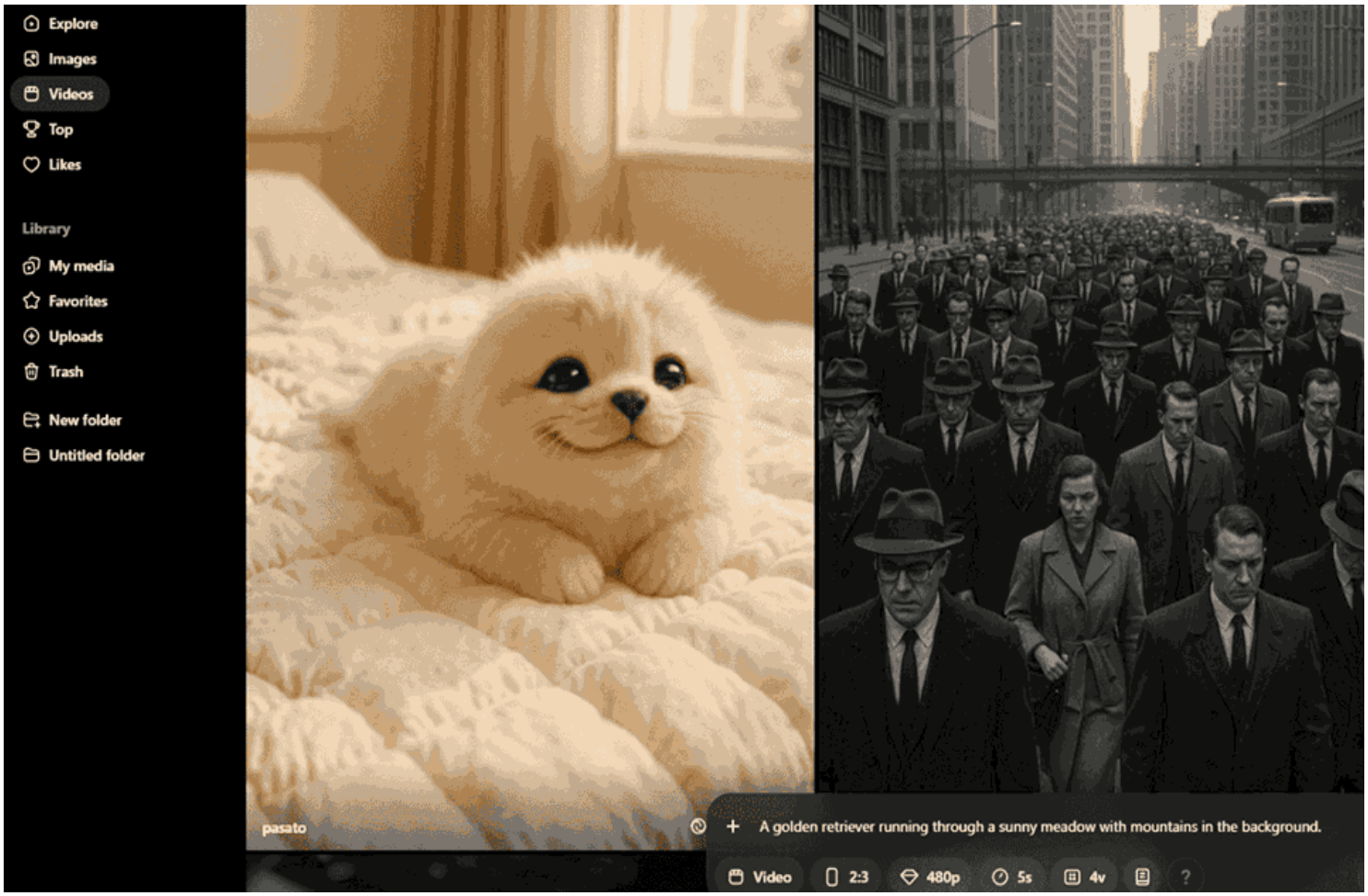


Text Prompt

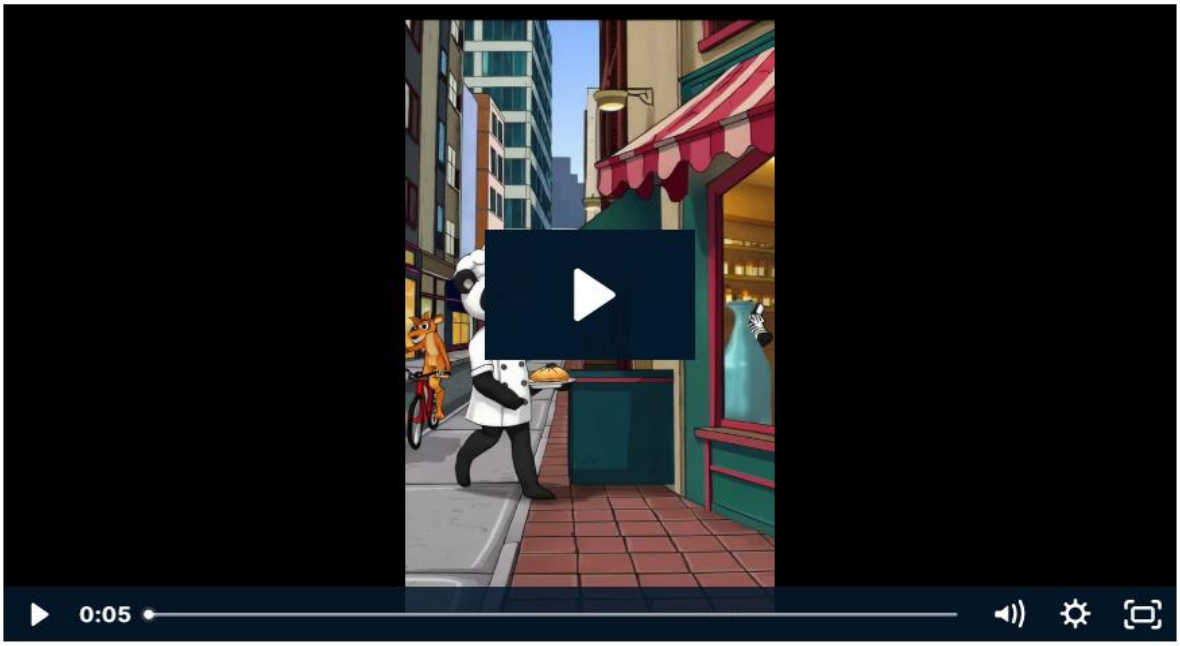
Text prompt engineering is vital in directing text-to-video models to produce videos that are visually striking while precisely meeting user specifications. This involves crafting detailed descriptions to instruct the model to effectively bridge the gap between human creativity and AI's execution capabilities. The prompts for Sora cover a wide range of scenarios. Recent works have shown how prompt engineering leverages model's natural language understanding ability to decode complex instructions and render them into cohesive, lively, and high-quality video narratives. The quality of prompt engineering depends on the careful selection of words, the specificity of the details provided, and comprehension of their impact on the model's output.

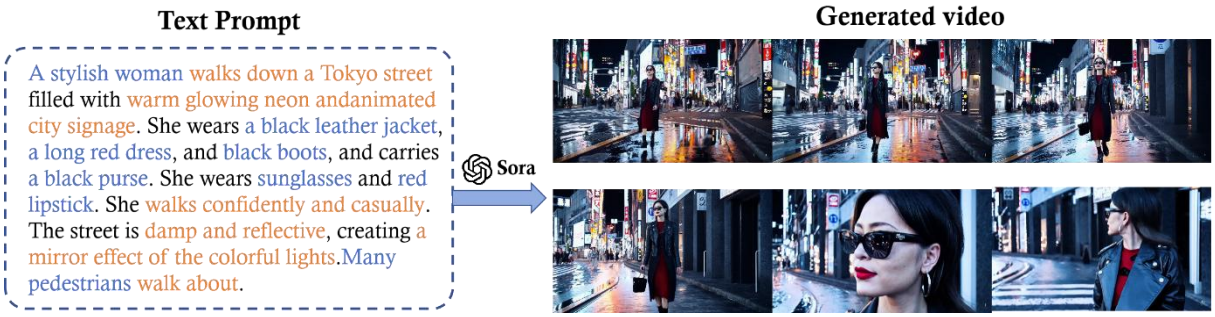
To create a video, simply describe it in the text prompt at the bottom of the screen.





Here's the result:

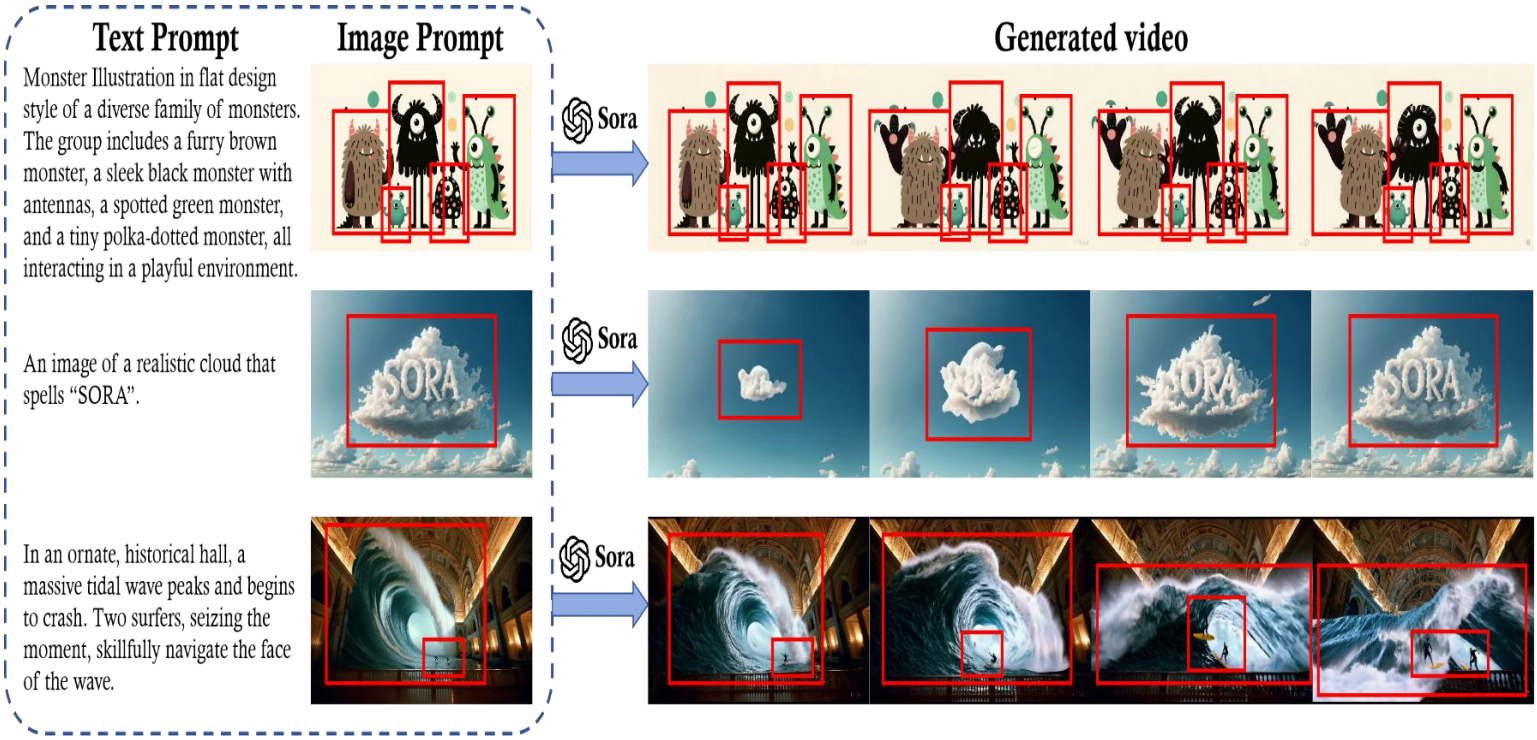




Example of SORA Text to Video Generation

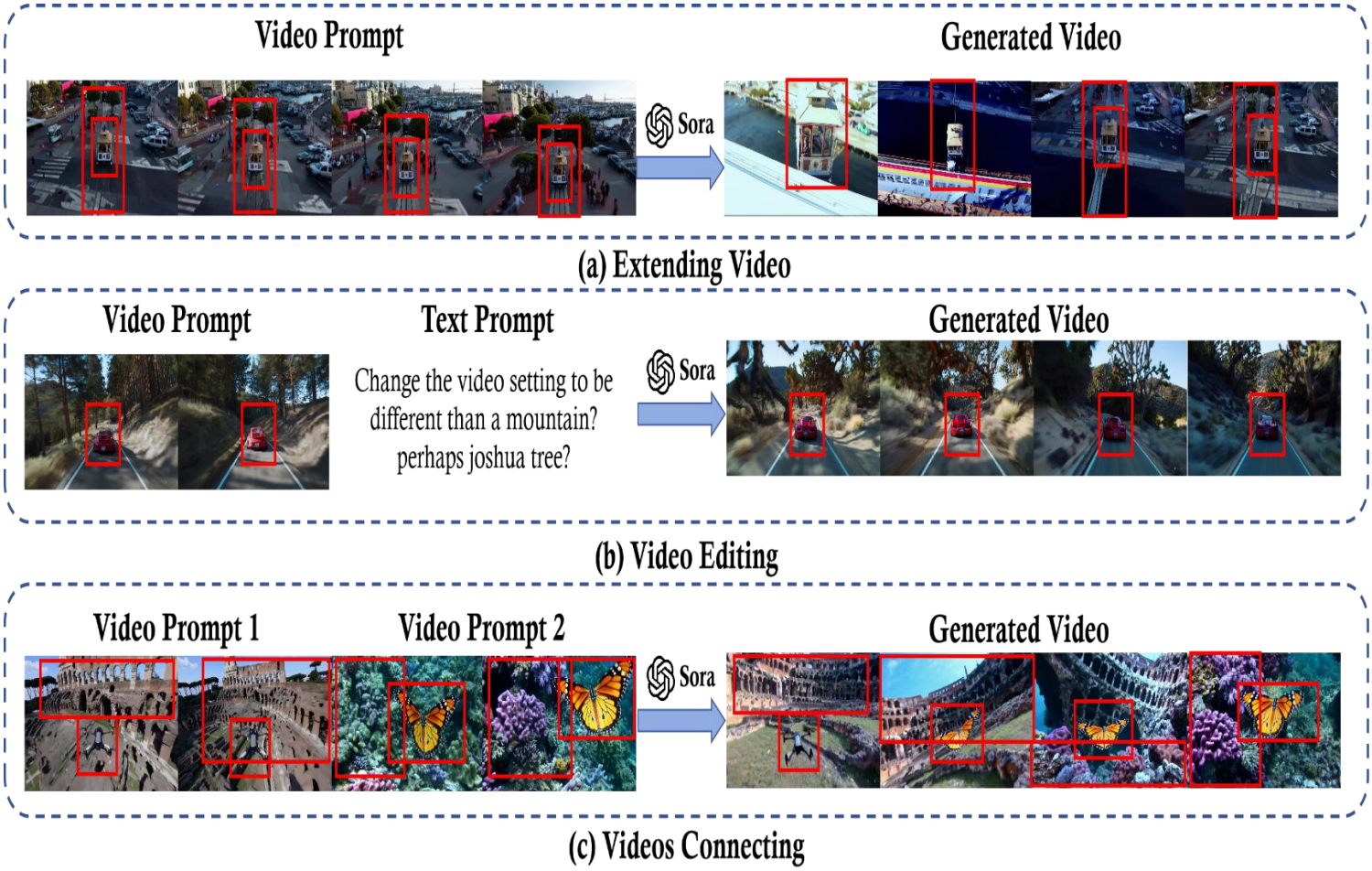
Image Prompt

An image prompt serves as a visual anchor for the to-be-generated video's content and other elements such as characters, setting, and mood. In addition, a text prompt can instruct the model to animate these elements by e.g., adding layers of movement, interaction, and narrative progression that bring the static image to life. The use of image prompts allows Sora to convert static images into dynamic, narrative-driven videos by leveraging both visual and textual information.



Video Prompt

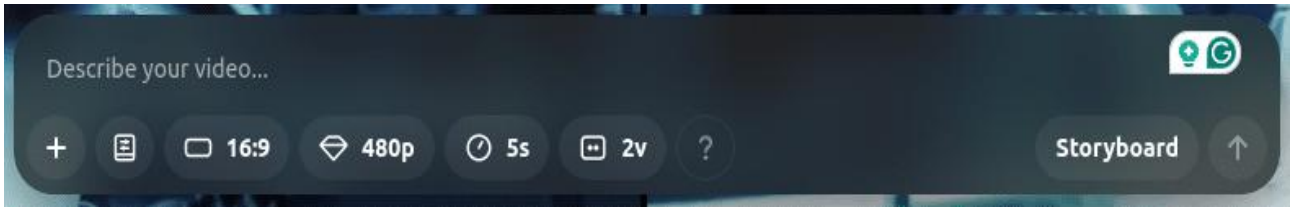
Video prompts can also be used for video generation as demonstrated in. Recent works that good video prompts need to be specific and flexible. This ensures that the model receives clear direction on specific objectives, like the portrayal of particular objects and visual themes, and also allows for imaginative variations in the final output. For example, in the video extension tasks, a prompt could specify the direction (forward or backward in time) and the context or theme of the extension. When performing video-to-video editing through video prompts, the model needs to clearly understand the desired transformation, such as changing the video's style, setting or atmosphere, or altering subtle aspects like lighting or mood.



How Does OpenAI's Sora Model Work ?

1. Input Processing: Before generating a video, Sora processes the input provided by the user. This input can be text, images, or existing videos.

A) Text-to-Video Input

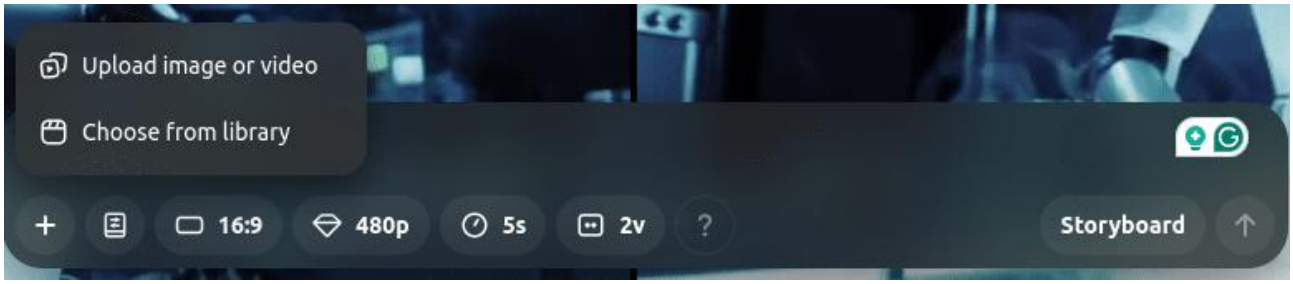


- The user provides a detailed text prompt describing the desired video scene.
- Sora's **natural language processing (NLP)** module interprets the text, breaking it down into key elements such as:
  - **Objects** (e.g., "a cat, a red car")
  - **Actions** (e.g., "running, jumping, swimming")
  - **Environment** (e.g., "a rainy street in Tokyo, a futuristic city")
  - **Artistic Style & Mood** (e.g., "cinematic, neon-lit, realistic")

Example:
A user inputs: "A golden retriever runs through a field of wildflowers with the sun setting in the background."

Sora identifies the dog, the field, the motion of running, and the lighting conditions of a sunset to generate a relevant scene.

B) Image-to-Video Input



- Users can upload an image as a starting point.
- Sora analyzes the image to extract:
  - Color palettes (e.g., warm tones of a sunset, vibrant city lights)
  - Textures & Materials (e.g., grass, water, fabric)
  - Perspective & Depth Information
- The AI then animates the image, adding movement and realistic details.

Example:
A still image of a beach at sunset can be turned into a video with waves crashing, birds flying, & the sun slowly setting.

C) Video-to-Video Input (Remixing & Enhancement)

- Users can upload a video that Sora will enhance, extend, or modify.
- The model analyzes movement, frame consistency, and transitions to maintain coherence.
- Users can request style changes, add objects, or modify backgrounds.

Example:
A daytime cityscape video can be transformed into a cyberpunk night scene with neon signs and rain reflections.

2. Latent Space Representation: Once the input is processed, Sora encodes it into a latent space. This step translates the input into a high-dimensional numerical format that captures key details like:

- Object relationships
- Motion patterns
- Color schemes and textures
- Perspective and depth

This process **compresses information** while preserving the structure needed for video generation.

Example:
The phrase “a futuristic car speeding through a neon-lit highway” is transformed into a numerical format that helps the AI generate consistent video frames.

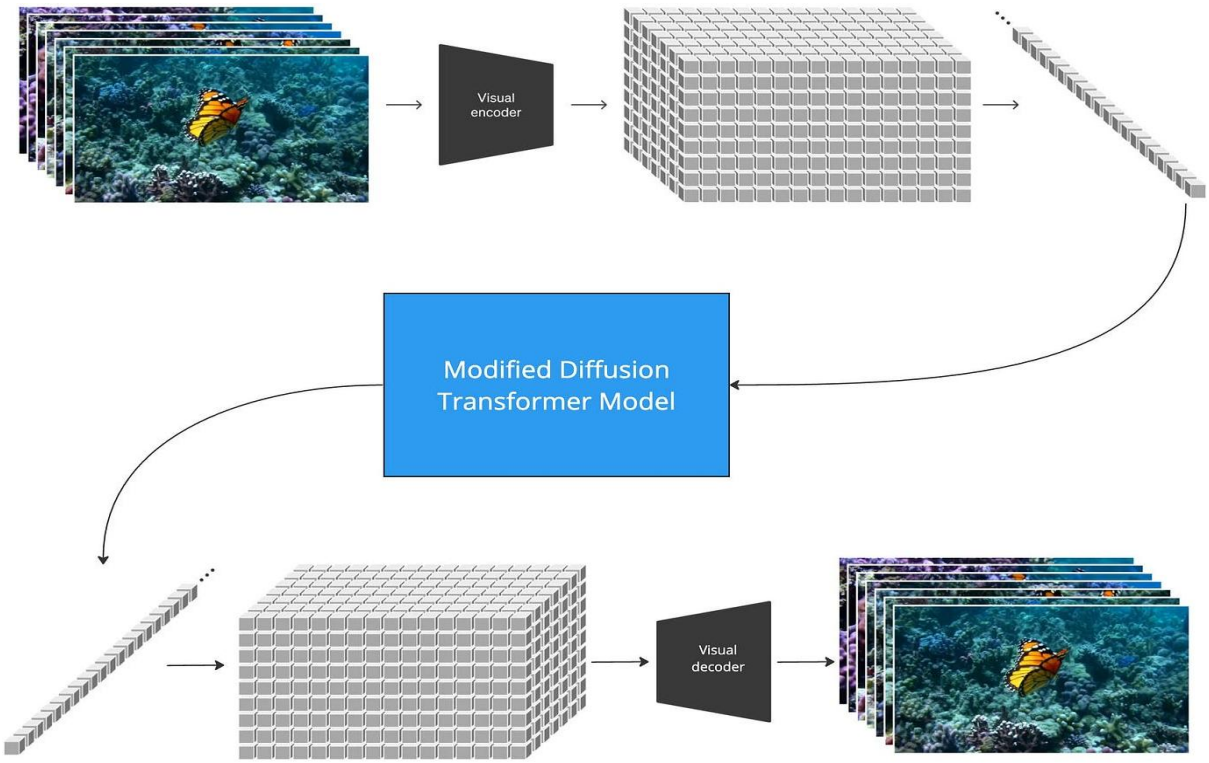
3. Diffusion Model Processing: Sora uses diffusion models to generate video frames from scratch. This involves:

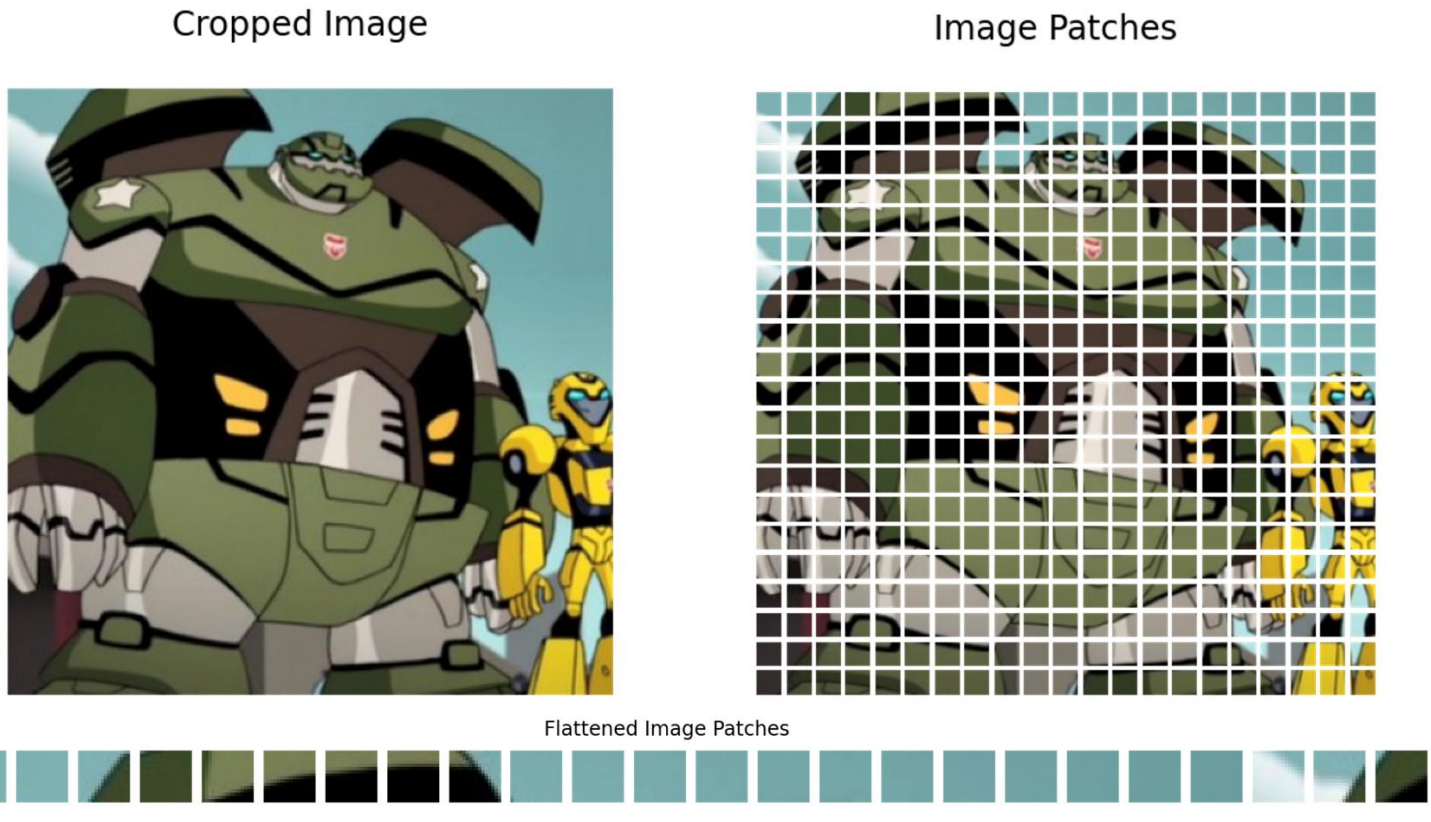
A) Noise Addition (Reverse Engineering the Image)

- The model starts with random noise (similar to static on a TV screen).
- It gradually removes the noise while shaping the pixels to match the prompt.

B) Iterative Refinement

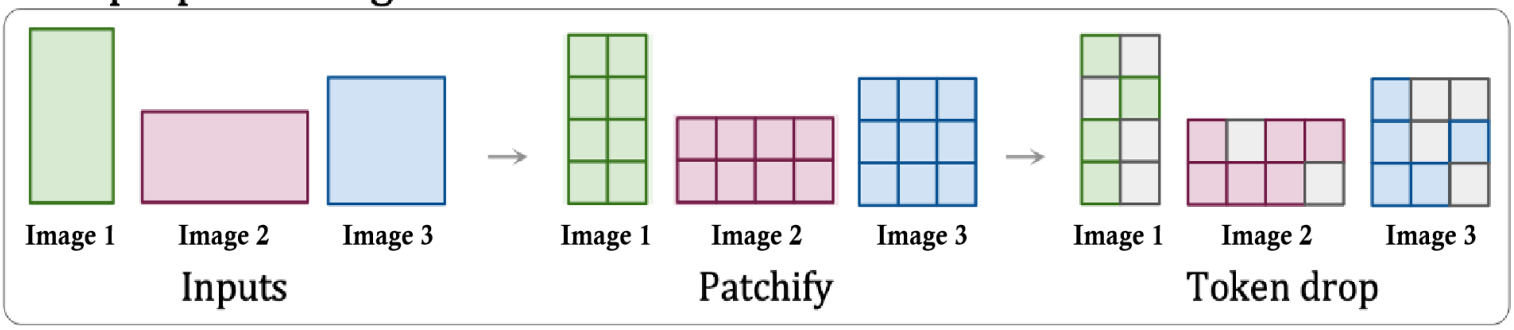
- Through multiple steps, the AI adds details, enhances textures, & improves clarity.
- The process ensures temporal consistency, meaning objects and actions remain smooth across frames.







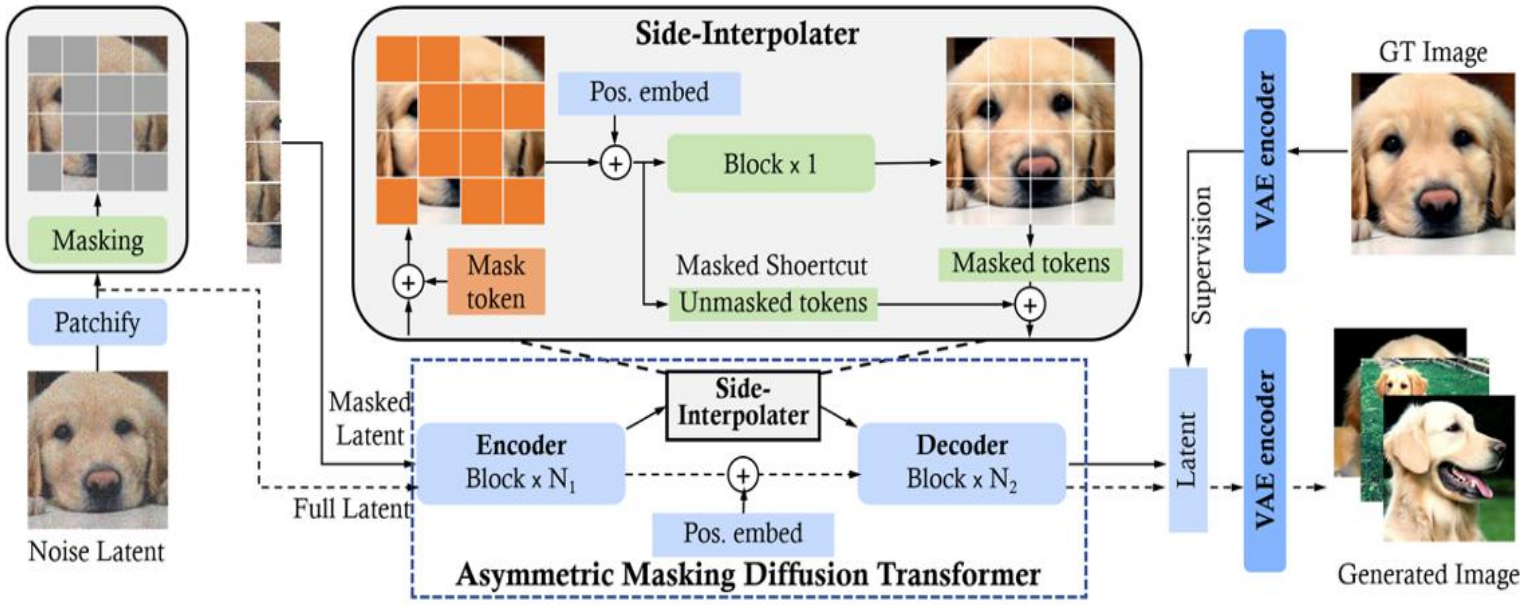
Data preprocessing



Example:

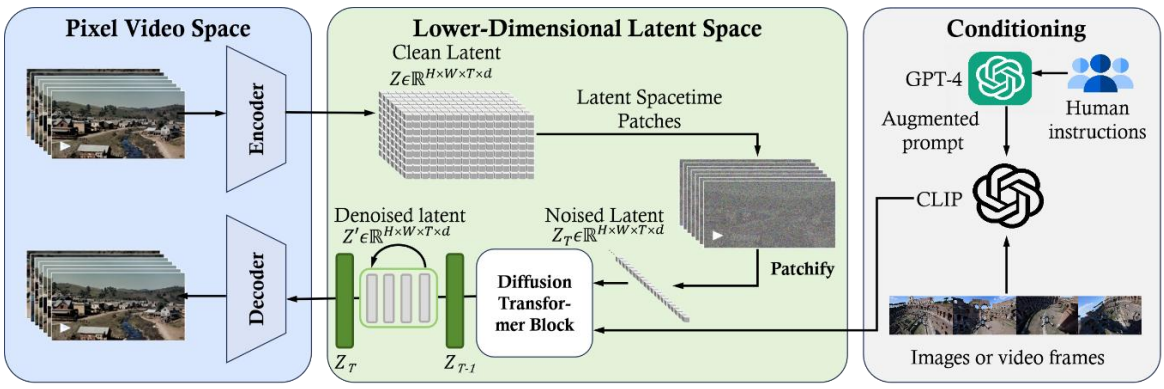
For the golden retriever running in a field, Sora ensures:

- The dog's fur flows naturally with the wind.
- The shadows move consistently as the sun sets.
- The background remains steady, avoiding glitches.



4. Transformer Model for Temporal Consistency: Unlike static image generators, video AI must handle motion. Sora integrates transformer-based architectures to ensure:

- Consistent object placement (so the same cat doesn't change shape in different frames).
- Realistic motion physics (like the way hair moves in the wind).
- Frame coherence (so there's no flickering or weird jumps).



Sora achieves this by analyzing:

- Sequences of frames to understand movement.
- Attention mechanisms that focus on important elements like a person's face, a moving car, or flowing water.

Example:
For a video of a dancer performing on stage, Sora ensures:

- The outfit moves naturally with the dance.
- The stage lighting changes smoothly.
- The dancer's movements don't glitch between frames.

5. Video Synthesis and Output Generation: Once Sora refines the video, it assembles and enhances the final output.

A) Frame Assembly



- The AI combines multiple video frames into a smooth sequence.
- It adjusts frame rates (e.g., 30 FPS, 60 FPS) for high-quality motion.

B) Post-Processing

- Color correction & lighting adjustments for realism.
- Stabilization & sharpness enhancement for crisp details.
- Final resolution selection (HD, 4K, etc.).

Example:
A forest scene at dawn might undergo:

- Brighter contrast adjustments to match the early morning light.
- Smoother tree movements in the wind.
- Higher-resolution textures for added realism.

6. Content Moderation & Safety Features: Sora is designed with ethical considerations to prevent misuse. The model:

- Adds watermarks & metadata to indicate AI-generated content.
- Restricts highly realistic human deepfakes to prevent fraud.
- Monitor input prompts to block inappropriate content.

Example:
If someone tries to generate a fake video of a celebrity, Sora will block or alter the request to prevent misuse.
By following these steps, Sora creates high-quality, dynamic videos that surpasses the boundaries of AI-powered video generation.

Writing a High-Quality Prompt

Sora relies on detailed prompts for better accuracy and creativity:-

A) **Structure of a Good Prompt**

1. **Main Subject** – What the video is about.
2. **Actions & Motion** – What's happening in the scene.
3. **Background & Environment** – Where the scene takes place.
4. **Style & Mood** – Animation style, cinematic, realistic, etc.

Example:
"A robot chef in a futuristic kitchen preparing sushi. The scene is lit with cool blue neon lights, & steam rises from the dishes. The camera moves smoothly from a first-person perspective."

Tip:

- Be specific (mention details like time of day, colors, and movements).
- Avoid vague prompts like "A cool animation"—Sora needs clear instructions.

Customizing Video Settings

Once you submit a prompt, Sora provides options to customize the output:-

A) **Resolution & Frame Rate**

- Choose Standard (HD), 4K, or Cinematic Quality based on your needs.
- Select a frame rate (30 FPS for smooth motion, 60 FPS for ultra-fluid videos).

B) **Duration & Length**

- Most AI-generated videos have a default length (e.g., 5-15 seconds).
- If longer videos are supported, you can extend duration by generating additional frames.

C) **Motion Style** (if applicable)
Choose between:

- Realistic Animation
- Cartoon/Anime Style
- Cinematic Slow Motion

Example:

- If creating an action scene, select 60 FPS, cinematic lighting, and slow-motion effects.

Generating the Video

Once all settings are configured, click Generate Video:-

A) **Processing Time**

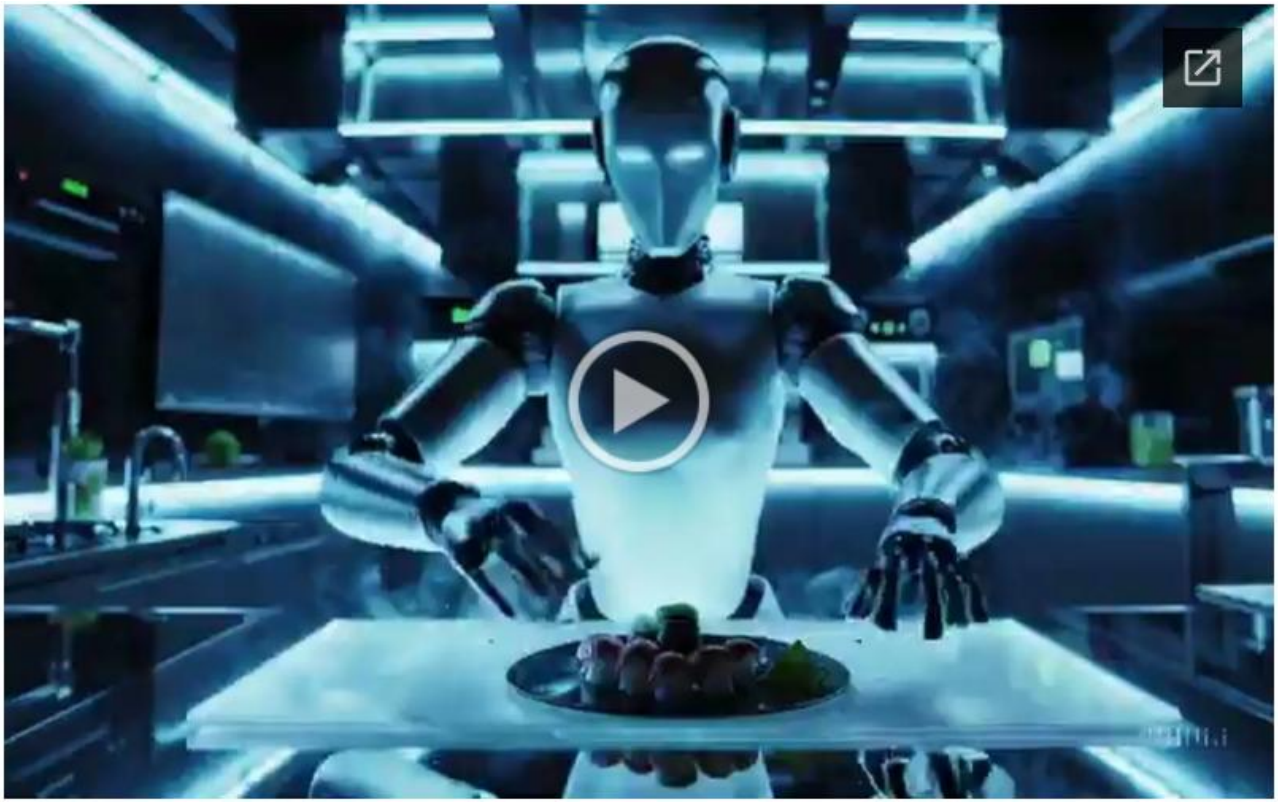
- Sora analyzes the input and begins generating frames.
- Depending on the complexity of the scene, processing may take a few minutes.

B) **Preview & Refinement**

- After generation, you'll see a low-resolution preview.
- If needed, you can make edits or adjust elements (e.g., colors, lighting, or movement speed).

Example:

- If a robot's movement is too fast, adjust the motion speed before final rendering.



Downloading & Sharing the Video

Once satisfied with the output, click Download to save your video.

A) **File Formats Available**

- MP4 (Standard video format)
- GIF (For short animations)
- MOV (For professional editing in tools like Adobe Premiere)

B) **Sharing Options**

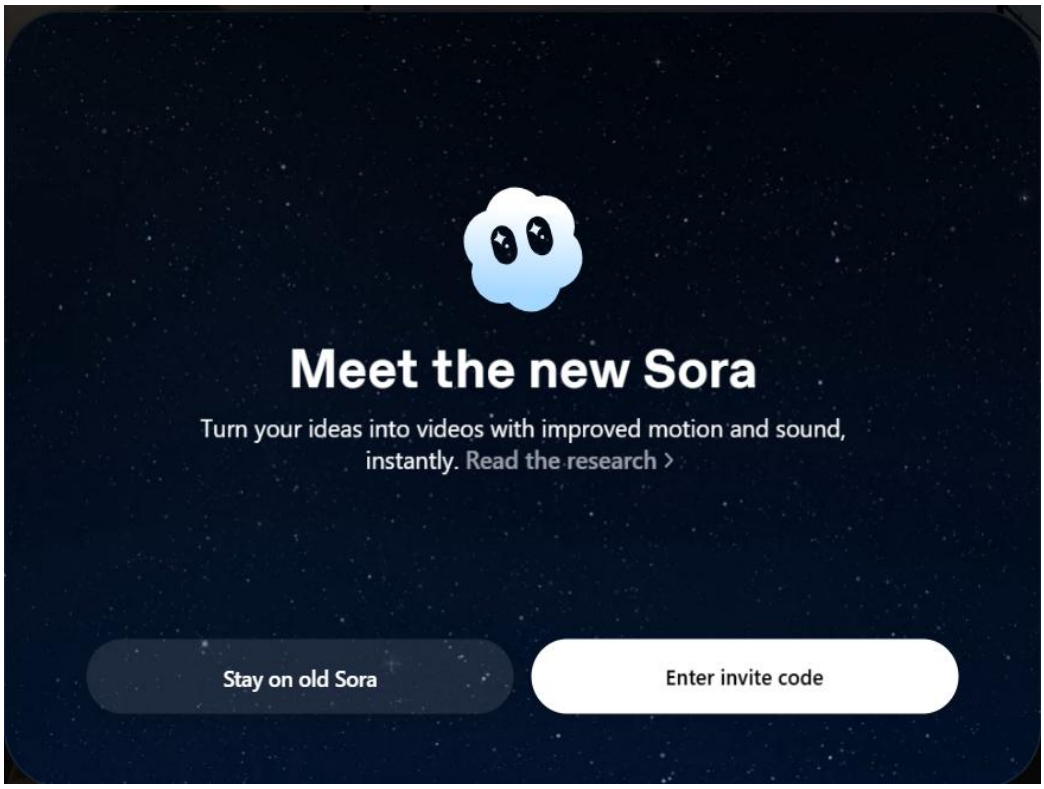
- Directly upload to social media (YouTube, Instagram, TikTok).
- Generate a sharable link for quick previews.

Pro Tip:

If using AI videos for marketing, add captions or voiceovers to make content more engaging.

SORA 2

OpenAI's **Sora 2** is a major leap in AI-driven video creation, building on the original *Sora* model from early 2024. Announced on September 30, 2025, Sora 2 combines an advanced video-generation model with a new social app, enabling users to create and share short, AI-generated videos with unprecedented realism.



SORA2 can be only accessed by invite code available to limited users via using the discord server of OpenAI. SORA2 was just recently launched, so, it is currently only available for IOS devices via App. However, in future, SORA2 will be available to all countries and will be available to all users to use without the need of invite code.

Technical Advancements Over Sora 1

Sora 2 represents a "GPT-3.5 moment for video," achieving feats that were impossible for the original Sora. Key technical improvements include:

- **Physical Realism and Motion**: Sora 2 dramatically improves world simulation. It respects physics and object permanence far better than Sora 1. For example, if a basketball misses the hoop, Sora 2 shows it rebounding off the backboard, instead of "cheating" reality by teleporting the ball into the hoop. Complex motions (Olympic gymnastics, backflips on a paddleboard) are modeled believably, a stark contrast to prior models that often morphed objects to fulfill prompts. This yields more natural movement and fewer visual artifacts, bringing AI video closer to real-world physics.

- **Higher Fidelity and Length:** Video quality and length have increased. While Sora 1 was a "rough but revolutionary" first step (some reports limited it to only ~6–8 second clips), Sora 2 can generate up to **10-second** videos with *4K-like detail*. (In practice, 10s is the in-app limit,

slightly exceeding Google's 8-second cap for Veo 3.) The model produces **sharper visuals, consistent frame continuity**, and handles diverse styles (cinematic, photorealistic, anime) with ease. Sora 2's largest configuration has even demonstrated generating longer clips internally (the original research noted up to *one minute* of high-definition video in a controlled setting, though such lengths are not yet public-facing due to compute constraints).

- **Synchronized Audio Generation:** A headline feature of Sora 2 is **integrated audio,** including dialogue, sound effects, and ambient sound perfectly synced to the video. This addresses a gap in Sora 1 (which produced silent or music-only videos). In Sora 2, characters' lips move in time with generated speech, and sound effects match on-screen actions. For example, a prompt of an old sailor gesturing at sea could yield crashing wave sounds and spoken lines in the sailor's voice. This audio-video sync marks a leap from "silent movie" AI to fully **multimodal storytelling**, greatly enhancing immersion.

- **Enhanced Prompt Control & Multi-Shot Scenes:** Sora 2 can follow **intricate, multi-part prompts** and maintain coherence across them. It handles **multiple shots within a single video**, persisting world state and character continuity from one scene to the next. For example, one can prompt a short narrative ("Scene 1: a rocket launches; Scene 2: astronauts floating in space…") and Sora 2 will honor the sequence, preserving details between cuts. This is a big improvement in **controllability,** Sora 2 listens to fine-grained instructions (camera angles, lighting, shot transitions) far better than its predecessor. Indeed, OpenAI demonstrated extremely detailed prompts specifying cinematographic parameters (lenses, shutter speed, color grading, etc.), which Sora 2 interprets to create polished, stylistically consistent clips. Such prompt fidelity was challenging for Sora 1, which often omitted details or jumbled complex scenes.

- **"Cameos" and Real-World Injection:** A standout new capability is the **Cameo** feature effectively personal deepfakes with consent. By uploading a short video of yourself (to verify identity and capture your likeness), you can inject *yourself or others* as realistic characters in any generated scene. Sora 2 accurately reproduces the person's appearance *and voice* in new contexts. For example, a user can record a cameo and then appear with correct face and speaking voice, as an adventurer in a Sora-generated fantasy or as the hero of an animated clip. This "upload yourself" feature required the model to master **character consistency** (keeping a person recognizable across scenes and motions). The Sora team "worked very hard" on this aspect, making it possible to drop a real human (or pet) into AI worlds with remarkable fidelity. This goes beyond static image prompting; it's a dynamic *video* likeness that moves and talks. Sora 1 lacked this capability entirely.

- **Audio-Visual Quality and Consistency:** Across the board, Sora 2 is more **stable and coherent**. Characters maintain their form from start to finish, and bizarre morphing or flickering is greatly reduced. The model better understands **object permanence and temporal consistency**, a benefit of training on larger video datasets. As a result, scenes in Sora 2 don't suffer from the disappearing limbs or melting faces that earlier generators sometimes produced. OpenAI notes Sora 2 is far from perfect and still makes mistakes, but many errors now feel like "plausible" mistakes (e.g. a human stuntman stumbling) rather than jarring glitches. This improvement validates that scaling up video-focused neural nets (plus *post-training on huge video data*) leads to more lifelike results.

Conclusion: SORA was studied, understood and practiced successfully.